

Philosophy of Science
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Lecture 8

Quantitative and Qualitative Approaches

Motivation of this lecture

- International studies often involves...
 - reconstructing complex causal pathways in social structures
 - formulating and testing causal hypotheses
- You need to be familiar with...
 - quantitative and qualitative methods theorised by social scientists
 - their strengths and weaknesses
- This lecture will give you that critical awareness

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Plan of this lecture

- Quantitative and qualitative methods
 - Aims, approaches, typical forms
- Comparison on two axes
 - Direction of causal inference
 - Place in empirical cycle
- Case study research
 - Example of qualitative methodology
- Internal and external validity: trade-offs

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Reading for this lecture

- Gary Goertz and James Mahoney, *A Tale of Two Cultures: Contrasting Qualitative and Quantitative Paradigms*
 - Princeton University Press, 2012
- Please read:
 - Chapter 1, "Introduction", pp. 1–15
 - Chapter 3, "Causes-of-effects versus Effects-of-causes", pp. 41–50
- Available on Brightspace

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Reminder

- No lectures in the next two weeks
 - 15 April 2026: Humanities Career Event
 - 22 April 2026: lecturer absent
- Lecture 9
 - Date: 29 April 2026
 - Topic: Rationality

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Quantitative vs. qualitative

- Social world is very complex
 - Many instances and many variables
 - Two routes to simplification
- Quantitative methods
 - Many instances, few variables—"Large N "
- Qualitative methods
 - Few instances, many variables—"Small N "
- Please note:
 - Not to be confused with the nomothetic vs. idiographic distinction

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Quantitative methods

- Aim
 - To perform causal inference by means of cross-case analysis
- Approach
 - Collect (quasi-)numerical data on relatively few variables from many instances
 - Analyse these data with statistical tools
- Typical forms
 - Interventional studies, experimental and quasi-experimental studies

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Qualitative methods

- Aim
 - To reconstruct causal pathways by means of within-case analysis
- Approach
 - Conduct in-depth study of relatively many variables in relatively few instances
- Typical forms
 - Observational studies, case studies

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Example: sociology of friendship

- . Typical quantitative research question
 - . How do friendship networks affect health and wellbeing outcomes?
- . Method
 - . Pick a wide pool of subjects with known friendship networks
 - . Gather numerical data on outcomes
- . In essence...
 - . Experiment that treats friendship as cause

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Friendship continued

- . Typical qualitative research question
 - . How do people form friendships?
- . Method
 - . Interview people with many and few friends
 - . Gather testimony on how they meet people, meanings of friendship, obstacles to forming relationships
- . In essence...
 - . Case study that treats friendship as effect

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Strengths and weaknesses

- . We now discuss trade-offs between quantitative and qualitative methods
 - . On two interrelated axes
- . Direction of causal inference
 - . Forward inference vs. reverse inference
- . Place in empirical cycle
 - . Hypothesis generation vs. hypothesis test

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Direction of causal inference

- . Both qualitative and quantitative methods can ground causal claims...
 - . But they tend to work in opposite directions
- . Qualitative methods
 - . Reverse causal inference: reasoning from effects to causes
- . Quantitative methods
 - . Forward causal inference: reasoning from causes to effects

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Direction: qualitative methods

- Qualitative researchers usually...
 - start from a real-world case in which a certain effect is apparent
 - and inquire what caused that effect
- They study causal relations between...
 - an already observed outcome
 - and hypothesised earlier interventions
- Reverse causal inference
 - Reasoning from effects to causes
- Example: case study method—see later

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Direction: quantitative methods

- Quantitative researchers usually...
 - know that a particular cause has operated in the set of instances under study
 - and gather data about the effect of that cause in the population
- They study...
 - consequences of a given intervention in a system
- Forward causal inference
 - Reasoning from causes to effects

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Quantitative methods: example

- Clinical drug trial compares outcomes of an intervention T for...
 - subjects that receive T
 - members of a control group
- Aim of the method:
 - To gauge the (average) effect of T on subjects
 - Not suited to confirm that T is the cause of the outcome for any particular subject
- Example of experimental study

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Place in empirical cycle

- Empirical cycle in science consists of various phases, including:
 - Hypothesis generation
 - Hypothesis test
- Example: Karl Popper's falsificationism
 - Context of discovery vs. context of justification—see lecture 5, sheet 21
- Each phase requires appropriate methods

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Place: qualitative methods 1

- . Qualitative methods are well suited to hypothesis generation
 - . In-depth knowledge of a few cases permits reconstruction of causal pathways
 - . Basis for tentative conjectures about relevant causal factors
 - . Heuristic value in exploratory research, awaiting test against a larger sample
- . Example
 - . Case study research

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Place: qualitative methods 2

- . Qualitative methods are less suited to hypothesis test
 - . Single cases are a poor basis for judging the validity of a general hypothesis
 - . Vulnerable to criticism of cherry-picking
- . Even falsification is scarcely applicable
 - . A single case might refute a hypothesis...
 - . but only that *A* is a necessary or sufficient condition for *B*—see lecture 5, sheet 29
 - . Rare in social science

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Place: quantitative methods

- . Quantitative methods are well suited to hypothesis test
 - . A good hypothesis will predict a certain correlation in quantitative data
 - . An experimental design is able to test that prediction effectively
- . They are not well suited to hypothesis generation
 - . Collecting quantitative data on relatively few variables provides a poor basis for reconstructing causal pathways

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Summary: qualitative methods

- . Good for:
 - . Reverse causal inference: reasoning from effects to causes
 - . Reconstructing complex causal pathways
 - . Generating hypotheses
- . Weaknesses:
 - . Danger of selection bias
 - . Difficulty of generalising findings

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Summary: quantitative methods

- Good for:
 - Forward causal inference: reasoning from causes to effects
 - Testing generalisations about relationships between variables
- Weaknesses:
 - Poor basis for reconstructing causal pathways
 - Unable to ascertain which causes produced the observed effects, or how they did so

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Overview of trade-offs

	Qualitative methods	Quantitative methods
Reverse causal inference Hypothesis formulation	✓	
Forward causal inference Hypothesis test		✓

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Concluding reflections

- Labels “qual”/“quant” are misleading
 - Whether a method uses quantitative or qualitative data is not the decisive issue
 - The two axes that we have explored characterise the distinction more deeply
- No hint of hierarchy or competition
 - Are qualitative methods “less scientific”?
 - Certainly not: both sets of methods play essential and irreducible roles in science

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Case study method—what is it?

- Case study research is...
 - intensive study of a single case
 - with the purpose to shed light on a larger class of cases
- Typical procedure:
 - Note an outcome in the case under exam
 - Try to ascertain what causal factor acting in the case was responsible for that outcome
- Instance of qualitative method

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Functions of case studies

- Case studies lend themselves more to...
 - hypothesis generation and explorative research...
 - than to hypothesis test, confirmation or disconfirmation
- A single-unit study...
 - allows us to ventilate multiple hypotheses
 - but we are unlikely to adopt or reject a hypothesis on the basis of it

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Who uses case study research?

- Case study research is widespread in...
 - History, sociology, political science, law, anthropology, psychology, economics
- Some disciplines are based almost entirely on case studies
 - Psychoanalysis: Sigmund Freud's clinical histories
 - Business studies: casework in MBA programmes
- Possible reason: theoretical weaknesses

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Case studies: illustration 1

- Frédéric Le Play (1806–1882)
 - French sociologist, pioneer of case study research
- Investigated the stresses acting on families in the era of industrialisation
 - Assigned fieldworkers to live with a family, gathering data on its attitudes, social life, and economic activities
- Reverse causal inference
 - Reasoning from effects to causes

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Case studies: illustration 2

- Robert S. Lynd and Helen M. Lynd
 - *Middletown: A Study in Contemporary American Culture*. New York, 1929
 - Study of the town of Muncie, Indiana
- Division between working and business class explains...

what one does all day long throughout one's life; whom one marries; when one gets up in the morning; whether one belongs to the Holy Roller or Presbyterian church; or drives a Ford or a Buick (pp. 23–24)

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Case studies: illustration 3

- Theda Skocpol's study of the French, Russian, and Chinese revolutions
 - Aim: to understand revolutions in general
 - *States and Social Revolutions: A Comparative Analysis of France, Russia and China*. Cambridge, 1979
- What makes a single-unit study a "case study" is...
 - the aim to understand a larger class
- What about representativeness?

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Internal and external validity 1

- Internal validity
 - Degree to which a study's findings are warranted for the case studied
- External validity
 - Degree to which a study's conclusions are generalisable
- Trade-off between forms of validity
 - Measures taken to enhance the warrant of findings (internal validity) tend to limit their generalisability (external validity)
 - and vice versa

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Internal and external validity 2

- Qualitative methods tend to score relatively high on internal validity
 - It is usually easier to establish a finding pertaining to a single case than a more general claim
- ... but less high on external validity
 - A case study may be unrepresentative of a wider population
 - Example: using a case study of France to try to understand nation-states

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Some complications

- Methodology of social science is more complex than described here
- For example, most present-day writers emphasise...
 - productiveness of integrating qualitative and quantitative approaches
- More sophisticated approaches
 - Mixed methods, triangulation of methods

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Recapitulation

- Two approaches in social science
 - Quantitative and qualitative methods
 - Comparison on two axes: direction of causal inference and place in empirical cycle
- Case study research
 - Example of qualitative methodology
- Internal and external validity
 - Trade-off between warrant and generalisability of findings